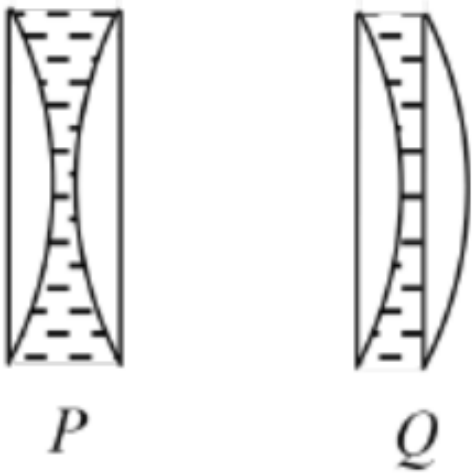
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A liquid of refractive index 1.33 is placed between two identical plano-convex lenses, with refractive index 1.50. Two possible arrangements, P and Q are shown. The system is



(A) divergent in P, convergent in Q

(B) convergent in P, divergent in Q

(C) convergent in both

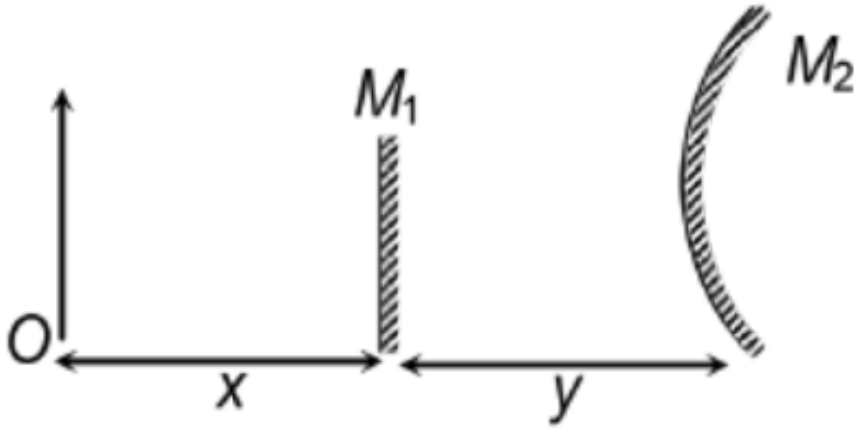
(D) divergent in both

Q.2

An object O is placed in front of a small plane M_1 and a large convex mirror M_2 of focal length f. The distance between O and M_1 is x, and the distance between M_1 and M_2 is y. The images of O formed

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by M_1 and M_2 coincide. The magnitude of f is



(A) $\frac{x^2-y^2}{2y}$

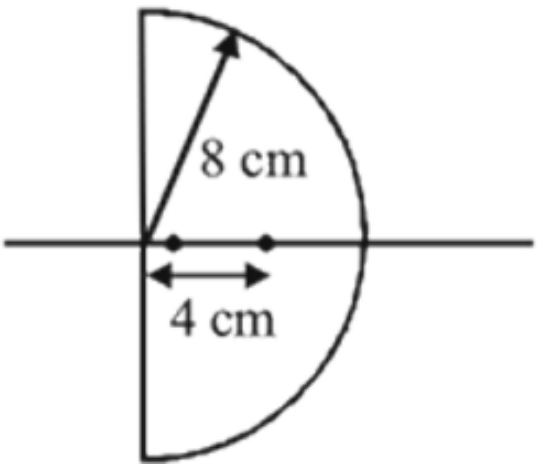
(B) $\frac{x^2+y^2}{2y}$

(C) $x-y$

(D) $\frac{x^2+y^2}{x-y}$

Q.3

A plastic hemisphere has a radius of curvature of $8\ \text{cm}$ and an index of refraction of 1.6. On the axis halfway between the plane surface and the spherical one $(4\ \text{cm}$ from each) is a small object O. The distance between the two images when viewed along the axis from the two sides of the hemisphere is approximately



(A) $1.0\ \text{cm}$

(B) $1.5\ \text{cm}$

(C) $3.75\ \text{cm}$

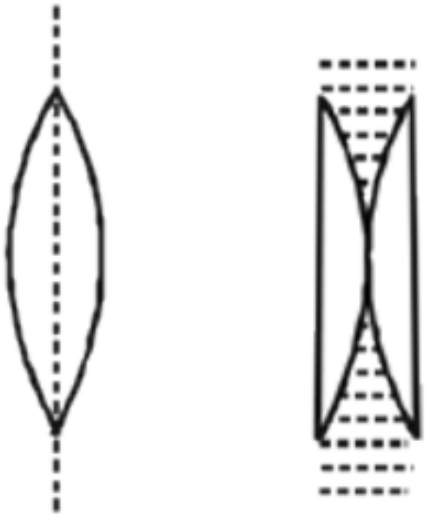
(D) $2.5\ \text{cm}$

Q.4

A thin convex lens of focal length $10\ \text{cm}$ and refractive index 1.5 is cut vertically into two equal pieces.

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They are placed as shown with a liquid of refractive index 3 between them. What is the focal length of the combination?



(A) -10 cm.

(B) $-10/4$ cm.

(C) $-10/3$ cm.

(D) None of these

Q.5

The following data are given for a crown glass prism ; refractive index for violet light $n_v = 1.521$ refractive index for red light $n_r = 1.510$ refractive index for yellow light $n_y = 1.515$ Then the dispersive power of a parallel glass slab made of the same material is

(A) 0.01

(B) 0.03

(C) 0

(D) 0.02

Q.6

Two beams of red and violet colours are made to pass separately through a prism (angle of the prism is 60°). In the position of minimum deviation, the angle of refraction will be

(A) 30° for both the colours

(B) greater for the violet colour

(C) greater for the red colour

(D) equal but not 30° for both the colours

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Q.7

For an optical fibre consists of a core (refractive index $= \mu_1$) and cladding (refractive index $= \mu_2$):

(A) $\mu_1 > \mu_2$

(B) $\mu_1 < \mu_2$

(C) $\mu_1 = \mu_2$

(D) $\frac{\mu_1}{\mu_2} = \frac{1}{2}$

Q.8

The image of an illuminated square is obtained on a screen with the help of a converging lens. The distance of the square from the lens is $40\ \text{cm}$. The area of the image is 9 times that of the square. The focal length of the lens is :

(A) $36\ \text{cm}$

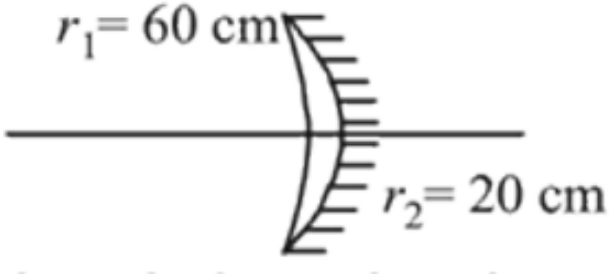
(B) $27\ \text{cm}$

(C) $60\ \text{cm}$

(D) $30\ \text{cm}$

Q.9

Convex surface of thin concavo - convex lens of refractive index 1.5 is silvered as shown. A small object is kept in air at $30\ \text{cm}$ left of the lens on its principal axis. The distance of the final image is



(A) $20\ \text{cm}$

(B) $30\ \text{cm}$

(C) $10\ \text{cm}$

(D) $15\ \text{cm}$

Q.10

Rays of light from Sun falls on a biconvex lens of focal length f and the circular image of Sun of radius r is formed on the focal plane of the lens. Then

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(A) area of image is πr^2 and area is directly proportional of f

(B) area of image is πr^2 and area is directly proportional to f^2

(C) intensity of image increases if f is increased

(D) if lower half of the lens is covered with black paper area will become half

Q.11

If the refractive index of the material of a prism is $\cot \frac{A}{2}$ and the angle of prism is A, then angle of minimum deviation is

(A) $\pi - 2A$

(B) $\pi - A$

(C) $\frac{\pi}{2} - 2A$

(D) $\frac{\pi}{2} - A$

Q.12

A telescope has an objective of focal length 100 cm and an eyepiece of focal length 5 cm. What is the magnifying power of the telescope when the final image is formed at the least distance of distinct vision ?

(A) 20

(B) 24

(C) 28

(D) 32

Q.13

An object at 2.4 m in front of a lens forms a sharp image on a film 12 cm behind the lens. A glass plate 1 cm thick, of refractive index 1.50 is interposed between lens and film with its plane faces parallel to film. At what distance (from lens) should object shifted to be in sharp focus of film?

(A) 7.2 m

(B) 2.4 m

(C) 3.2 m

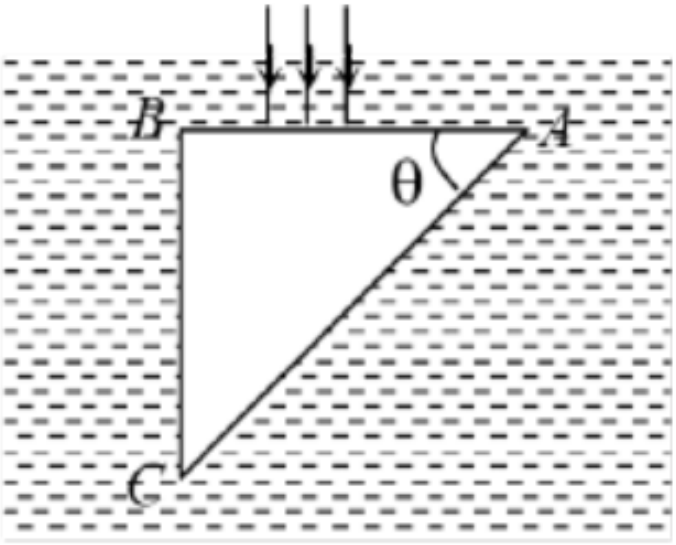
(D) 5.6 m

Q.14

A glass prism of refractive index 1.5 is immersed in water $(\mu = 4/3)$. A light beam

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incident nomally on the face AB is totally reflected to reach the face BC, if



(A) $\sin\theta \geq \frac{8}{9}$

(B) $\sin\theta \leq \frac{2}{3}$

(C) $\sin\theta = \frac{4}{5}$

(D) $\frac{2}{3} < \sin\theta < \frac{8}{9}$

Q.15

A diminished image of an object is to be obtained on a screen $1.0\ \text{m}$ from it. This can be achieved by appropriately placing

(A) a concave mirror of suitable focal length

(B) a convex mirror of suitable focal length

(C) a convex lens of focal length less than $0.25\ \text{m}$

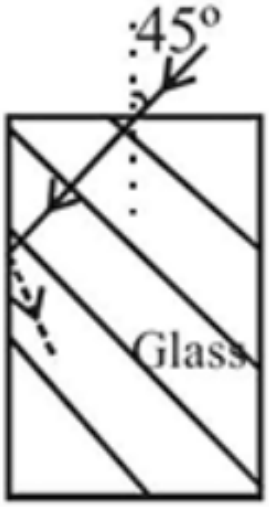
(D) a concave lens of suitable focal length

Q.16

A light ray falls on a rectangular glass slab as shown. The index of refraction of the glass, if total internal

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reflection is to occur at the vertical face, is



(A) $\sqrt{3/2}$

(B) $\frac{(\sqrt{3}+1)}{2}$

(C) $\frac{(\sqrt{2}+1)}{2}$

(D) $\sqrt{5}/2$

Q.17

An equilateral prism produces an angle of deviation 60° when the angle of incidence is 60°. The angle of deviation produced when the ray is incident normal to the surface is

(A) 30°

(B) 45°

(C) 60°

(D) can't be determined

Q.18

If light travels a distance x in time t_1 sec in air and 10x distance in time t_2 in a certain medium, then find the critical angle of the medium.

(A) $\sin^{-1}\left(\frac{20t_1}{t_2}\right)$

(B) $\sin^{-1}\left(\frac{10t_1}{t_2}\right)$

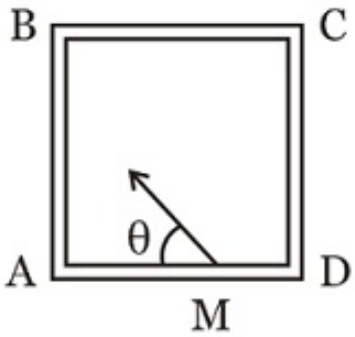
(C) $\sin^{-1}\left(\frac{t_1}{t_2}\right)$

(D) $\sin^{-1}\left(\frac{t_2}{10t_1}\right)$

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Q.19

AB, BC, CD and DA are identical mirror used to from a square as shown in the figure(Top view). A ray starting from the midpoint M of AD undergoes two reflections before reaching the corner D. Find angle θ.



(A) $\tan^{-1}(0.75)$

(B) $\cot^{-1}(0.75)$

(C) $\sin^{-1}(0.75)$

(D) $\cos^{-1}(0.75)$

Q.20

A plano-convex lens is joined to a concave lens of the same focal length. The effect of this will be

(A) the focus shifts to infinity.

(B) the focal point shifts towards the lens by a small distance.

(C) the focal point shifts away from the lens by a small distance.

(D) the focus remains undisturbed.

Answers & Solutions

Q.1 Answer:

convergent in both

Solution:

for arrangement P,

$\frac{1}{f_{eq}} = \frac{1}{2R} + \frac{1}{2R} - \frac{2}{3R} = \frac{1}{3R}$

$(R =$ **radius of curvature** $)$

For arrangement Q,

$\frac{1}{f_{eq}} = \frac{1}{2R} + \frac{1}{2R} - \frac{1}{3R} = \frac{2}{3R}$

$\therefore$

Q.2 Answer:

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$\frac{x^2-y^2}{2y}$

Solution:

Due to M_1, an image is formed at a distance x from M_1, i.e. at a distance $(x-y)$ behind M_2.

Thus, for M_2

$u=-(x+y), v=x-y$

Use $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$\therefore$

Q.3 Answer:

2.5 cm

Solution:

Distance of image from the plane surface is as follows:

$x_1=\frac{4}{1.6}=2.5 \text{ cm}\left(d_{\text{app}}=\frac{d_{\text{actual}}}{\mu}\right)$

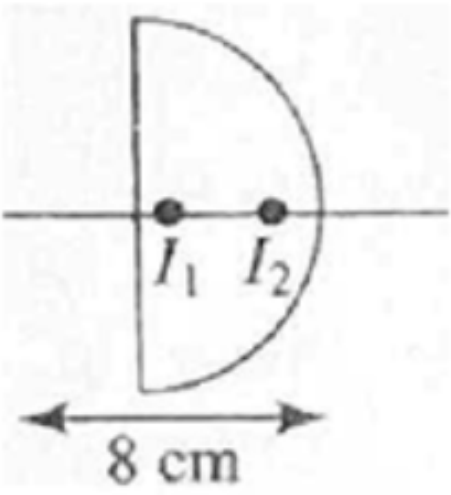
For the curved side

$$\frac{1.6}{4}+\frac{1}{x_2}=\frac{1-1.6}{8}$$

$\therefore \quad x_2 \approx -3.0 \text{ cm}$

The minus sign means the image is on the object side.

$\therefore \quad \mathrm{I_1I_2}=(8-2.5-3.0)\text{cm}=2.5 \text{ cm}$



Q.4 Answer:

$-10/3$ cm.

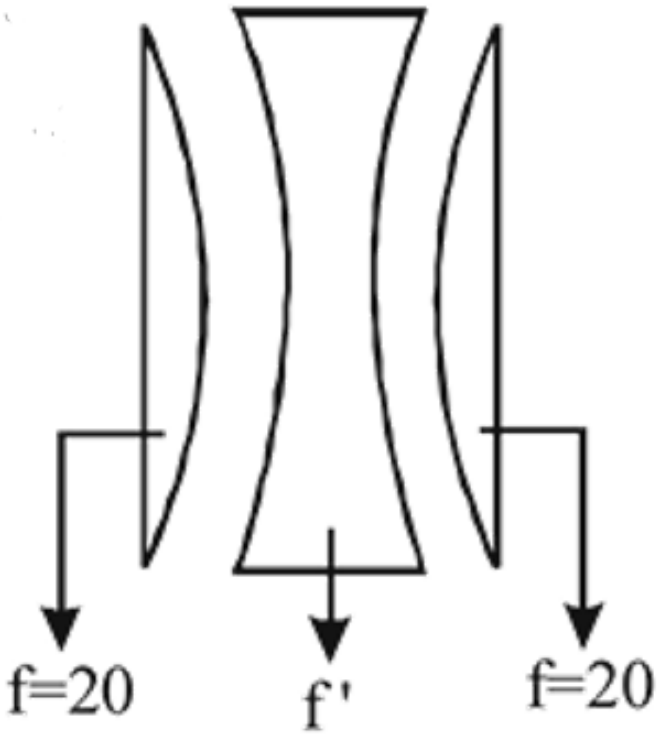
Solution:

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$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \ ;$$
$$R = 10 \text{ cm.}$$
$$\frac{1}{f'} = (3 - 1)\left(\frac{1}{-10} - \frac{1}{10}\right),$$

$f' = -\frac{10}{4}$

$\frac{1}{f_{eq}} = \frac{1}{20} - \frac{4}{10} + \frac{1}{20} = \frac{2}{20} - \frac{4}{10} \Rightarrow f_{eq} = -\frac{10}{3}$



Q.5 Answer:

0.02

Solution:

Dispersive power $(\omega) = \frac{n_v - n_r}{n_y - 1} = \frac{1.521 - 1.510}{1.515 - 1} = 0.02$

Q.6 Answer:

30° for both the colours

Solution:

For minimum deviation the ray in the prism is parallel to the base of the prism. This condition does not depend on the colour (or wave length) of incident radiation. So in both the cases, by geometry, $\mathrm{r} = 30^\circ$**. So** (a) **is correct option.**

Q.7 Answer:

$\mu_1 > \mu_2$

Solution:

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No solution available

Q.8 Answer:

30 cm

Solution:

If side of object square $= \ell$ **and side of image square** $= \ell'$ **From question,** $\frac{\ell'^2}{\ell} = 9$ **or** $\frac{\ell'}{\ell} = 3$

i.e., magnification $\mathrm{m} = 3 \quad \mathrm{u} = -40$ cm

$\mathrm{V} = 3 \times 40 = 120$ cm$\mathrm{f} = ?$

From formula, $\frac{1}{\mathrm{V}} - \frac{1}{\mathrm{u}} = \frac{1}{\mathrm{f}}$

$\frac{1}{120} - \frac{1}{-40} = \frac{1}{\mathrm{f}}$ **or,** $\frac{1}{\mathrm{f}} = \frac{1}{120} + \frac{1}{40} = \frac{1+3}{120}$

$\therefore \mathrm{f} = 30$ cm

Q.9 Answer:

10 cm

Solution:

Power of combinations $P = 2(\mu - 1)\left(\frac{1}{-60} - \frac{1}{-20}\right) + \left(\frac{-1}{-20/2}\right)$

$P = \frac{2}{15} = \frac{-1}{f}$

$\therefore$ **Resultant focal length** $f = -\frac{15}{2}$

$\therefore$ **From mirror formula** $\frac{1}{-30} + \frac{1}{v} = \frac{1}{f}$

Solving $v = -10$ cm.

Q.10 Answer:

area of image is $\pi \mathrm{r}^2$ and area is directly proportional to f^2

Solution:

No solution available

Q.11 Answer:

$\pi - 2\,\mathrm{A}$

Solution:

(a) We have,

$\mu = \frac{\sin\left(\frac{\mathrm{A}+\delta_\mathrm{m}}{2}\right)}{\sin\left(\frac{\mathrm{A}}{2}\right)}$

$\Rightarrow \cot\frac{A}{2} = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

or $\sin\frac{A}{2} \cdot \cot\frac{A}{2} = \sin\left(\frac{A+\delta_m}{2}\right)$

or $\sin\frac{A}{2} \cdot \frac{\cos\frac{A}{2}}{\sin\frac{A}{2}} = \sin\left(\frac{A+\delta_m}{2}\right)$

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or $\cos \frac{\mathrm{A}}{2}=\cos \left[\frac{\pi}{2}-\left(\frac{\mathrm{A}+\delta_{\mathrm{m}}}{2}\right)\right]$
$\Rightarrow \frac{\mathrm{A}}{2}=\frac{\pi}{2}-\left(\frac{\mathrm{A}+\delta_{\mathrm{m}}}{2}\right)$
or $\mathrm{A}=\pi-\mathrm{A}-\delta_{\mathrm{m}} \Rightarrow \delta_{\mathrm{m}}=\pi-2 \mathrm{~A}$.

Q.12 Answer:

24

Solution:

$\mathrm{f}_0=100 \mathrm{~cm}, \mathrm{f}_{\mathrm{e}}=5 \mathrm{~cm}$
When final image is formed at least distance of distinct vision (d), then $\mathrm{M}=\frac{\mathrm{f}_0}{\mathrm{f}_{\mathrm{e}}}\left(1+\frac{\mathrm{f}_{\mathrm{e}}}{\mathrm{d}}\right)=\frac{100}{5}\left(1+\frac{5}{25}\right) \quad[\because \mathrm{D}=25 \mathrm{~cm}]$
$\mathrm{M}=20 \times \frac{6}{5}=24$

Q.13 Answer:

5.6 m

Solution:

The focal length of the lens $\frac{1}{f}=\frac{1}{v}-\frac{1}{u}=\frac{1}{12}+\frac{1}{240}=\frac{20+1}{240}=\frac{21}{240}$
$f=\frac{240}{21} \mathrm{~cm}$
Shift $=t\left(1-\frac{1}{\mu}\right)$
$=1\left(1-\frac{1}{3/2}\right)=1 \times \frac{1}{3}$
Now $v'=12-\frac{1}{3}=\frac{35}{3} \mathrm{~cm}$
Now the object be distance u'. $\frac{1}{\mathrm{u}'}=\frac{3}{35}-\frac{21}{240}=\frac{1}{5}\left[\frac{3}{7}-\frac{21}{48}\right]$
$\frac{1}{\mathrm{u}'}=\frac{1}{5}\left[\frac{48-49}{7 \times 16}\right]$
$u'=-7 \times 16 \times 5=-560 \mathrm{~cm}=-5.6 \mathrm{~m}$

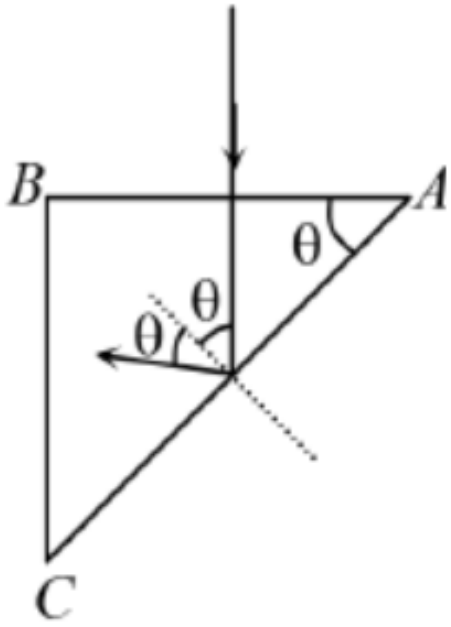
Q.14 Answer:

$\sin \theta \geq \frac{8}{9}$

Solution:

$\sin \theta>\sin \theta_C=\frac{1}{\mu}$
$\sin \theta>\frac{1}{\frac{3/2}{4/3}}, \sin \theta>\frac{8}{9}$

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Q.15 Answer:

a convex lens of focal length less than $0.25\ \mathrm{m}$

Solution:

(c)

Q.16 Answer:

$\sqrt{3/2}$

Solution:

For point A, $_a\mu_g = \frac{\sin 45^\circ}{\sin r}$

$\Rightarrow \sin r = \frac{1}{\sqrt{2}_a\mu_g}$

For point B, $\sin(90^\circ - r) = {}_g\mu_a$ **where,**

$(90^\circ - r)$ **is critical angle.**

$\therefore \cos r = {}_g\mu_a = \frac{1}{{}_a\mu_g}$

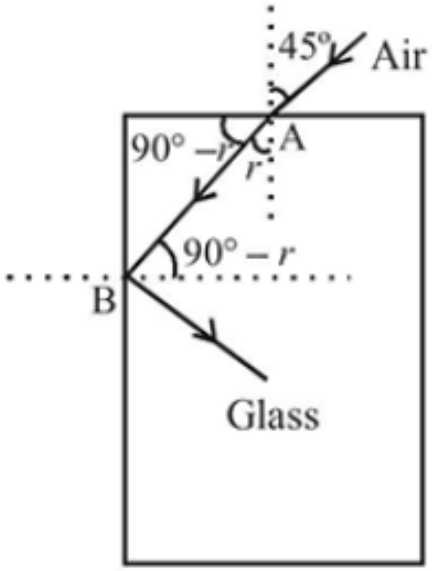
$\Rightarrow {}_a\mu_g = \frac{1}{\cos r}$

$= \frac{1}{\sqrt{1-\sin^2 r}} = \frac{1}{\sqrt{1-\frac{1}{2_a\mu_g^2}}}$

$\Rightarrow {}_a\mu_g{}^2 = \frac{1}{1-\frac{1}{2_a\mu_g^2}} = \frac{2_a\mu_g^2}{2_a\mu_g^2-1}$

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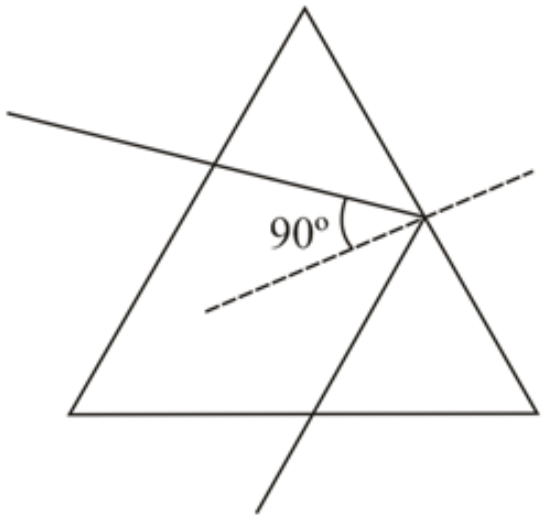
$\Rightarrow {}_a\mu_g^2 - 1 = 2 \Rightarrow a\mu_g = \sqrt{\frac{3}{2}}$



Q.17 Answer:

60°

Solution:



TIR **will take place**

Q.18 Answer:

$\sin^{-1}\frac{10t_1}{t_2}$

Solution:

$$\mu = \frac{\frac{x}{t_1}}{\frac{10x}{t_2}} = \frac{t_2}{10t_1}$$

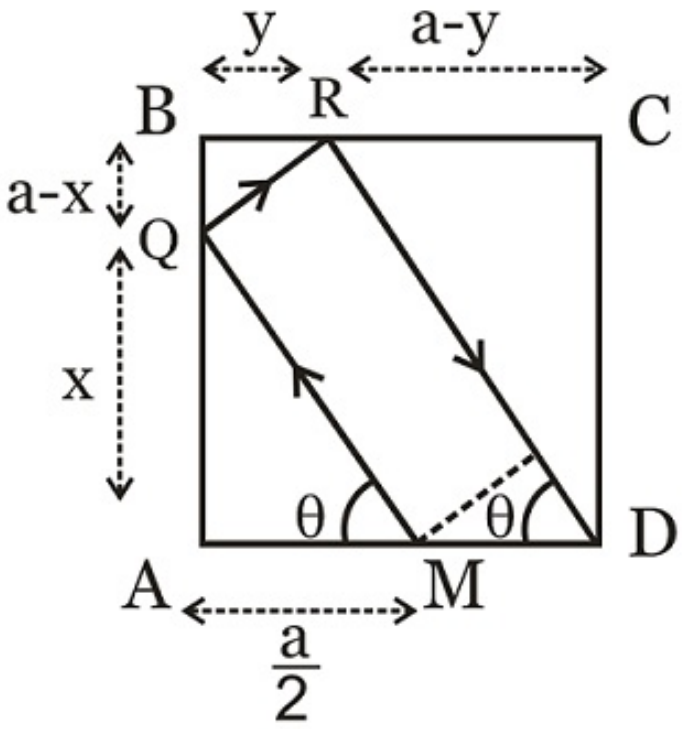
$$\sin\theta_C = \frac{1}{t_2 / 10t_1} = \frac{10t_1}{t_2} \Rightarrow \theta_C = \sin^{-1}\frac{10t_1}{t_2}$$

Q.19 Answer:

$\cot^{-1}(0.75)$

Solution:

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The situation shown in the figure

$\tan\theta = \frac{2x}{a} = \frac{a - x}{y} = \frac{a}{a - y}$

On solving

$\cot\theta = \frac{3}{4}$

$\Rightarrow \cot^{-1}0.75$

Q.20 Answer:

the focus shifts to infinity.

Solution:

The combined focal length of plano-convex lens

$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

Given, $f_1 = \infty\left(for\ plane\ surface\right),\ f_2 = fsay$

$\therefore \frac{1}{F} = \frac{1}{\infty} + \frac{1}{f}$

$\Rightarrow F = f$

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Now when concave lens of same focal length is joined to first lens , then combined focal length

$$\frac{1}{F'} = \frac{1}{F_1} + \frac{1}{F_2}$$

$$= \frac{1}{f} - \frac{1}{f} \qquad \left(\because F_1 = f,\ F_2 = -f\right)$$

$$= 0$$

$$F' = \infty$$

Thus, the image can be focused on infinity (∞) or focus shifts to infinity.

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